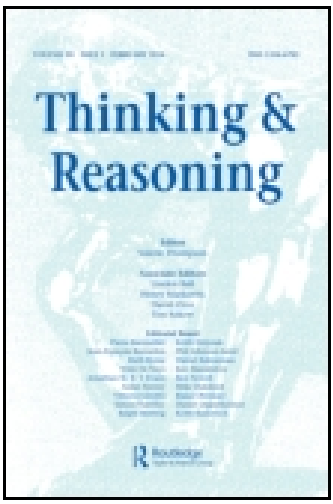
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Publisher: Routledge
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Thinking & Reasoning
Publication details, including instructions for authors and subscription information:
http://www.tandfonline.com/loi/ptar20

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Published online: 30 Oct 2013.

To cite this article: Jonathan St. B. T. Evans (2014) Two minds rationality, Thinking & Reasoning, 20:2, 129-146, DOI: 10.1080/13546783.2013.845605

To link to this article: http://dx.doi.org/10.1080/13546783.2013.845605

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Thinking & Reasoning, 2014
Vol. 20, No. 2, 129–146, http://dx.doi.org/10.1080/13546783.2013.845605

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Two minds rationality

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I argue that views of human rationality are strongly affected by the adoption of a two minds theory in which humans have an old mind which evolved early and shares many features of animal cognition, as well as new mind which evolved later and is distinctively developed in humans. Both minds have a form of instrumental rationality—striving for the attainment of goals—but by very different mechanisms. The old mind relies on a combination of evolution and experiential learning, and is therefore driven entirely by repeating behaviours which succeeded in the past. The new mind, however, permits the solution of novel problems by reasoning about the future, enabling consequential decision making. I suggest that the concept of *epistemic rationality*—striving for true knowledge—can only usefully be applied to the new mind with its access to explicit knowledge and beliefs. I also suggest that we commonly interpret behaviour as irrational when the old mind conflicts with the new and frustrates the goals of the conscious person.

Keywords: Rationality; Two minds; Dual process theory; Evolution.

The Great Rationality Debate (Stanovich, 2011b) has been running in the psychology of reasoning and decision making for the last 30 years and more, following the claim of Cohen (1981) that no amount of experimental evidence of bias and error could demonstrate that human beings are irrational. There are a number of conceptions of rationality, which I shall discuss later, but I have made it clear over the years that I have no commitment to *normative rationality*, the idea that people *ought* to conform to standard systems such as binary logic and probability theory in their everyday reasoning (most recently Elqayam & Evans, 2011; but see also Evans, 1993, 2007a; Evans & Over, 1996). However, I will address here more broadly based definitions of epistemic and instrumental rationality, defined below.

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I would like to thank Shira Elqayam, Keith Stanovich and an anonymous referee for a critical reading of an earlier draft of this paper.

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My particular purpose in this paper is to consider rationality within the framework of the two minds theory (Evans, 2010), an overarching account of what are popularly known as dual process and dual system theories of higher cognition (Evans, 2008; Kahneman, 2011). At the core of dual process theories is the idea that there are two qualitatively distinct forms of cognitive processing:

Type 1 (intuitive): fast, automatic, high capacity
Type 2 (reflective): slow, controlled, low capacity

This basic distinction is to be found in centuries of philosophical writing and throughout the history of psychology, with authors often unaware of the similar writings of others (Frankish & Evans, 2009). In contemporary psychology there are multiple dual process theories to be found in the literatures on learning, social cognition, reasoning, decision making, and other topics. The features attributed to Type 1 and Type 2 processes are extended in many of these accounts so that, for example, Type 1 processing may be described as unconscious, associative, parallel, belief based, and biased, and Type 2 processing as conscious, rule-based, sequential, abstract, and normative. As these features accumulated, a "received" version (Evans, 2012; Evans & Stanovich, 2013) has somehow emerged in which all of these attributes are seen to be part of *the* theory of dual processing, even though no one author has proposed the full list. This has allowed critics to attack a "straw man" version belonging to no one (Evans & Stanovich, 2013). In fact, many of these features have been disputed in the recent literature; for example, I have been pointing out for some time that it is gross simplification to attribute biases to Type 1 processes and correct performance to Type 2 processing (e.g., Evans, 2007a, 2008); there are dual process theories that assume both processes to be strongly influenced by belief (Verschueren, Schaeken, & d'Ydewalle, 2005) and so on.

The two minds theory is really a development of the idea of *dual systems* underlying Type 1 and 2 processing (e.g., Epstein, 1994; Evans & Over, 1996; Reber, 1993; Stanovich, 1999, 2004). The various accounts of two systems, dubbed Systems 1 and 2 by Stanovich (1999), had in common the idea that System 1 was evolutionarily ancient, sharing key features with animal cognition, while System 2 was evolutionarily recent and uniquely human (Evans, 2003). However, I later came to have great misgivings about the two systems terminology (Evans, 2006) and both I and Stanovich and stopped using it in our recent writing (see Evans & Stanovich, 2013). There were a number of reasons for this, including the following: (a) it was evident that the brain uses multiple systems for cognitive processing; hence there was a danger that people would think exactly two systems were in play; (b) it was apparent that some processing attributed to System 1 was not based in older

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regions of the brain; (c) modular cognition, which had been placed on System 1 list, was in some cases essential in supporting Type 2 thinking (see below). However, it is important to understand that, in abandoning the two systems terminology, I have not moved away from the idea that there are older, animal-like forms of cognition which combine with more recent and distinctively human kinds. This is what the two minds theory is about.

TWO MINDS IN ONE BRAIN

The essence of the two minds theory (Evans, 2010) is that humans have both an old mind, which resembles the cognitive systems of higher non-human animals, and a new mind which is uniquely developed in humans. As a result, we have two ways of knowing and deciding. The theory makes the strong assumption that there are distinct systems of learning and memory within the two minds (for evolutionary argument in support of this, see Reber, 1993; Sherry & Schacter, 1987). There is an ancient learning system shared with other animals which allows associative and procedural learning to occur, shaped by the environment in which we operate. Of course, it may be more developed in humans than in some of the other species, but it is still founded in old, subcortical regions of the brain, such as the basal ganglia (Eichenbaum & Cohen, 2001). This together with basic emotions provides the foundation for the old mind.

The new mind, by contrast, draws on explicit knowledge systems such as episodic and semantic memory which are processed through working memory (Baddeley, 2007), also known as controlled attention (Engle, 2002). These functions are not uniquely human and it was an error in earlier forms of dual system theories to describe System 2 as unique to human beings (Toates, 2004). Higher animals, as well as humans, have higher-order and controlled forms of cognition that can be recruited when habitual or instinctive responses are failing to deal with the situation (Toates, 2006). However, this system is rudimentary by comparison in all non-human animals. The new mind evolved distinctively in modern human beings, for we are of course unique in our engagement in science, engineering, and literature, as well as in our ability to design our environments to suit ourselves, rather than having to rely on the slow processes of natural selection to evolve adaptations. Nevertheless, it is a basic tenet of two minds theory that the new mind did not replace the old but co-exists with it. Hence we can experience *two minds conflict* when, for example, we become compulsive gamblers or suffer from severe phobias. In such cases the old mind takes control and may completely frustrate the goals of the new mind.

In general, the two minds cooperate such that the old mind takes responsibility for that which is habitual and requires no deliberative thought to achieve, while the new mind has the ability to deal with novelty and solve

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tasks that require controlled attention. Despite claims that a unimodel provides a more parsimonious account of human cognition (e.g., Kruglanski & Gigerenzer, 2011), I believe this two minds architecture to be an efficient proposal. Working memory capacity is severely limited so it makes sense for us to go with habitual and familiar responses most of the time, using learning and action systems that make minimal demands on this precious resource. However, it is equally important for us to be able to solve novel and demanding problems with which such implicit mechanisms cannot cope. In such cases we apply deliberative thinking which taxes working memory resources and is related both to the load of competing tasks and the cognitive ability of the participants, a very close correlate of working memory capacity (Colom, Rebollo, Palacios, Juan-Espinosa, & Kyllonen, 2004). Hence the characteristics typically associated with Type 2 processing—e.g., low capacity, slow, serial—seem to correspond with new mind activity, as do its claimed functional benefits of enabling hypothetical thinking and cognitive decoupling (Evans & Over, 1996; Evans & Stanovich, 2013; Stanovich, 2011b). But the story is not simply that of Type 2 processing in the new mind or System 2, and Type 1 processing in the old mind or System 1 (Evans, 2009). This is because most of the processing in the new mind is also Type 1 in the sense of being autonomous, rapid, and preconscious. For example, the prior knowledge that is required for solving a problem in a given context just effortlessly "comes to mind" depending on its relevance to the context. So the processes that retrieve this knowledge and post it in working memory clearly fit the definition of Type 1 and not Type 2 processing.

The problem is illustrated by consideration of modular cognition, in the sense defined by Fodor (1983). Such modules are self-contained innate elements of the mind which are dedicated to processing information in particular ways for limited purposes. Obvious examples lie within the visual system which has dedicated systems for detecting brightness, edges, movement etc. The processes involved are fast, automatic, mandatory, and parallel, hence fitting the criteria for Type 1 processing. But attributing modular cognition to System 1 (e.g., Stanovich, 1999) is not compatible with a two minds theory in which modules play a critical foundation for the *new* mind. To be sure, there are ancient modules within, for example, our perceptual systems but these provide inputs for both old and new mind learning. Also, there are some modules, more recently evolved, which seem distinctively human and essential to the new mind. For example, without the aid of linguistic representations and the meta-representational[1] power provided by theory of mind, it is difficult to see how the flexible and higher-order forms of cognition associated with human beings could operate (Evans, 2010).

[1] For arguments that meta-representation, linked to theory of mind, is a modular function unique to humans, see Evans (2010), Chapter 2.

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We should not rely on the distinction between conscious and unconscious processing, as both old and new minds have aspects that are conscious and unconscious (Evans, 2010, Ch. 7). The key difference is that the old mind operates through automated Type 1 systems that have mandatory outputs, whereas Type 2 systems are in some sense *volitional*: the new mind is capable of forming plans and carrying out intentions under controlled attention. Of course, we have many such plans of widely varying timescales and they lie mostly beneath our consciousness. Short-term plans mostly dominate our working memories—e.g., making lunch, answering an e-mail—but much longer-term plans such as maintaining a relationship or advancing a career are also tracked within the new mind and will demand access to working memory when the context requires it. Thus the short-term goal of writing a steaming response to an irritating e-mail may be overridden by the long-term plan of advancing our careers, when the e-mail in question was written by the boss. While multiple plans may lie dormant in the new mind, they do require access to a singular limited working memory resource to be explicitly advanced. By contrast, the modular, implicit, and procedural systems of the old mind can operate in parallel and without access to working memory, enabling us to move around and perform simple tasks while thinking about something entirely different.

An important class of dual process theories have a structure which is default-interventionist (Evans, 2007b; Evans & Stanovich, 2013). Default behaviours are produced by automated processes of the old mind but may be overridden by intervention from the new mind with its special abilities for dealing with novel problems by explicit reasoning. Controlled attention is required when automated processes are not sufficient. We are all familiar with the example of routine driving during which we can talk to a passenger or plan our day's work ahead. But adverse weather or traffic conditions will compete for working memory space, forcing our attention to the driving itself and away that which previously occupied it. A form of intervention with controlled behaviour occurs also in higher animals when habitual or instinctive responding fails (Toates, 2006), although their capability for Type 2 thought is very limited by comparison with ours. Intervention is important when we consider how the two minds combine to provide effective rationality for individuals.

NOTIONS OF RATIONALITY

The psychology of reasoning that I joined around 1970 was unashamedly normativist. There was an agreed standard normative system—the binary logic of truth and falsity. The deduction paradigm (Evans, 2002) was designed to see whether people untrained in formal logic could nevertheless

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solve logical problems. This reflected a long philosophical tradition in which logical reasoning was equated with rationality (Henle, 1962). Early reviews of work in this field pointed again and again to the errors and biases that people exhibited on these tasks (Evans, 1982; Wason & Johnson-Laird, 1972), leading some authors (especially Wason) to make explicit claims that people were irrational. In the study of judgement and decision making the importation of game theory from economics (Edwards, 1977) provided another equally clear normative system; people's decisions should conform with maximising expected utility and the necessary probability judgements should in turn conform with the probability calculus. The classic work of Amos Tversky and Danny Kahneman undermined both of these systems, first by showing that judgements of probability were typically based on heuristics leading to cognitive biases (Kahneman, Slovic, & Tversky, 1982; Tversky & Kahneman, 1974) and later by providing an alternative account of how people combine risk with prospects of gain or loss in decision making (Kahneman & Tversky, 1979).

The apparent evidence for human irrationality in both of these fields of work was disputed by the philosopher Jonathan Cohen (1981) who adopted the position later dubbed as Panglossian (Stanovich, 1999) that experiments could never show people to be irrational. I (Evans, 1993) later classified his arguments into three categories which I called the interpretation, normative system, and external validity problems. The first was that participants might have interpreted the task differently from the way the experimenter intended; the second that they might use a different normative system; and the last that they might have been tricked by unrepresentative and artificial psychological experiments. Cohen's paper changed these fields permanently. Ever since then, authors working in this field have explicitly addressed the issue of rationality. For example, Stanovich (1999) attacked Cohen's Panglossianism, paying particular attention to the problem of alternative construals and alternative norms, and has continued regularly (Stanovich, 2004, 2009, 2011b) to address rationality in this later writing on reasoning, decision making, dual processes, and individual differences. Oaksford and Chater (1998, 2007) have pursued an alternative norms policy, arguing that people are largely rational when their reasoning is viewed from a Bayesian perspective. Evans and Over (1996) argued that many problems are caused by confusing normative rationality (conformity with the rules of a normative system) with instrumental rationality (acting in such a way as to achieve personal goals). More recently Elqayam and Evans (2011) argued that an obsession with normative rationality had damaged both the psychology of reasoning and decision making, leading to serious biases in the framing of research questions, design of studies, and interpretation of their findings.

Of most relevance to the current paper is to note that multiple notions of rationality have emerged from the philosophical and psychological

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literatures. In philosophy, a classical distinction is that between instrumental and epistemic rationality. The former concerns behaviour: an organism is rational when it acts in such a way as to achieve its goals. This idea is easily applied to non-human animals, as in the vast programme of research inspired by B. F. Skinner on operant conditioning or instrumental learning. Skinner's work marked the later stages of the behaviourist school of psychology. The simple idea was that behaviours are shaped by environment. They may be produced randomly to start with, but those that are rewarded will be repeated and those that are punished will be withheld. This "random generation, selective retention" algorithm is essentially the same as the Darwinian mechanism of natural selection during evolution. Whereas the latter enables the surviving individual to pass on their characteristics, mutating the species, the former allows behaviours to "evolve" in the lifetime of an individual. In both cases, the fitness of features or behaviours for their environment is the key driver.

Epistemic rationality would not have been recognised by Skinner or any other behaviourist, because it refers to knowledge. Philosophers of mind talk of belief-forming systems (Carruthers, 2006) whose epistemic rationality is classically described as *truth seeking*. In psychological terms, we can think of epistemic rationality as the propensity of learning systems to form accurate representations of the world around us. However, if epistemic rationality requires our belief systems to be truth seeking, then much study in social psychology shows that we are apparently irrational. People may form and maintain any number of beliefs which are unfounded or demonstrably false when subjected to strong social influence of the groups around them (see Evans, 2010, Ch. 6). Such research strongly suggests that beliefs serve other functions such as social cohesion and bonding. Within the domain of cognitive psychology, however, we can reasonable consider whether people form knowledge of the world which is appropriate to facilitate their cognitive goals.

Normative rationality is equated with instrumental rationality by some authors (Stanovich, 1999, 2011b) while explicitly separated from it by others (Elqayam & Evans, 2011; Evans & Over, 1996). Its definition, conforming with a normative system, is certainly different from that of instrumental rationality—acting so as to achieve one's goals. I prefer the separation because, while the aim of a normative theory might be to provide a recipe for instrumental rationality, there are often multiple normative theories for the same situation. Also, when a normative theory fails to account for behaviour, psychologists have the habit of proposing an alternative normative theory which does fit the behaviour observed, rendering it instrumentally rational by a circular argument (Elqayam & Evans, 2011). For the purposes of discussing two minds rationality, therefore, I will focus mostly on the notions of instrumental and epistemic rationality, while ignoring

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normative theory where possible. The idea of evolutionary rationality (Stanovich, 1999, 2004), reflecting the goals of the genes rather than the individual, is also important and I will return to it below.

Finally, I offer a distinction between objective and subjective views of rationality. The objective view is that behaviour is irrational if the individual fails to act efficiently to achieve goals defined by some external standard. This approach is clearly useful in education where, for example, we want individuals studying mathematics to find the solution deemed correct by mathematicians, rather than alternatives that they might find subjectively satisfying. We want, for example, to say that individuals giving the intuitively compelling answers on Frederick's (2005) Cognitive Reflection Test are in some sense *wrong*. The objective view is advanced by "Meliorist" authors such as Stanovich (1999, 2011b) whose objective is to improve rational thinking through education. Hence his insistence on the importance of normative rationality (Stanovich, 2011a). The subjective view is that an individual is rational if they act in such a way as to achieve their personal goals, whether or not this conforms to a normative theory. This view was advanced by Evans and Over (1996) in their distinction between personal and normative rationality. The subjective approach was also recently developed by Elqayam (2012) who terms this concept "grounded rationality". On this view an individual can only be viewed as irrational if they fail to achieve their personal goals and could have chosen to act otherwise.

TWO MINDS RATIONALITY

It may seem that we could simply differentiate two kinds of rationality associated with Type 1 and Type 2 processing. For example, we could say that a Type 1 process is instrumentally rational if it provides behavioural control which is adaptive, or perceptual representations that are fit for purpose. We could also say that Type 2 thinking is rational if, for example, it leads to accurate reasoning and problem solving. But a problem with this approach is that in practice both Type 1 and Type 2 processes are often involved in the same behaviours and may operate in cooperation or in conflict. Nor, as mentioned earlier, can we simply map Type 1 processing into the old mind and Type 2 into the new mind. I believe that the two minds theory provides a much richer framework within which we can understand two forms of rationality: one that we share with other animals and one that is distinctively human. I shall argue here that old mind rationality is driven by what succeeded in the past, and that new mind rationality reflects reasoning about the future. I shall also claim that, while both minds serve instrumental rationality at the level of the individual, epistemic rationality is essentially a concept restricted to the new mind, with its explicit knowledge and belief systems.

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OLD MIND RATIONALITY

Although the old mind in humans is more complex, it is helpful to think about the cognition of higher animals in order to understand the kind of rationality it provides. Animals have essentially two forms of "stimulus controlled" behaviour (Toates, 2006): instincts and learned behaviours. Instincts are innate, following fixed patterns and universal to the species. They reflect adaptation to past environments over a lengthy period of evolution. Even higher animals have instinctive behaviour patterns such as the hunting behaviour of cats and dogs. However, the higher the animal on the evolutionary tree, the less the reliance on instinct and the more behaviour tends to reflect individual learning. Higher animals, especially primates, may also have limited cultural transmission of knowledge, passed on from one generation to the next. However, the appearance of cultural differences in animals can be misleading. For example the orcas (killer whales) of the North Pacific are observed in two kinds of social group or pod: residents that stay in one place and transients that move around. They also have different pod sizes and favour different diet: fish for residents and sea mammals for transients. While the animals appear to be identical, DNA studies have now shown that the two groups are quite clearly differentiated genetically, effectively forming two sub-species (Hoelzel, Dahlheim, & Stern, 1998).

Darwinian evolution operates for the benefit of neither individuals nor species but rather the "selfish genes" (Dawkins, 1976) that have the power to replicate themselves through their animal vehicles. This is why it is important to distinguish between instrumental rationality which serves the goals of an individual and evolutionary rationality which serves the goals of the genes. Stanovich (2004) has argued that while the individual and genetic goal sets have much overlap, they can also diverge. It is in the interests of the genes that their vehicles survive long enough to reproduce and raise the next generation to the level of sexual maturity, but not much more.[2] The general instrumental learning system that animals (and humans) possess evidently helps to protect the individual, although the motives such as hunger, thirst, and lust are genetically programmed. In most animals, however, instinctive behaviours are mandatory no matter what their outcome for the individual.

Evolutionary psychology pursues the hypothesis that much human behaviour is also genetically programmed but has proved highly controversial, especially in its strongest form in which all behaviour is attributable to innate Darwinian cognitive modules (Barkow, Cosmides, & Tooby, 1992), a thesis strongly rejected by Fodor himself (e.g., Fodor, 2001). Of course, we do not need to support "massive modularity" to acknowledge that there are

[2] I am pleased to note, however, that human grandparents are quite useful and that there may be an evolutionary basis for them hanging around (Coall & Hertwig, 2010).

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innate modular systems underlying perception, language, and so on and which have been shaped by evolutionary processes. There is also an impressive range of evidence gathered by evolutionary psychologists to support the idea that much of our social behaviour, e.g., mating preferences, has some innate basis (Buss, 2004). For the current purposes this debate is not too important. Suffice it to say that the old mind in human beings reflects a combination of evolutionary and instrumental rationality, provided by genetic programming and individual learning respectively. The learning I am referring to here is of the type we might call implicit or procedural: we acquire habits and skills through these mechanisms.

What characterises old mind rationality is adaption to the past (Evans, 2010). We repeat behaviours that have been successful in the past history of our species (evolutionary rationality) or more commonly in the past history of our individual lives (instrumental rationality). In both cases we do what has worked before, our behaviour being shaped by the environment in which we evolved or that in which we have lived as individuals. What the old mind cannot do—and this is the big limitation of animal cognition—is to make decisions by reasoning about future consequences. That now leaves us with the question of whether epistemic rationality is a useful concept when talking about the old mind, and I rather think it is not. It is clear in evolutionary terms that epistemic rationality should subservient to instrumental rationality and not be an end in itself. The pursuit of knowledge should serve the goals of the individual or their genes.

For epistemic rationality to be a useful concept, it should be clearly differentiated from simply instrumental behaviour. However, in the case of old mind functions, I argue that the knowledge we have is implicit in the behavioural control mechanisms. To know something in a procedural sense is to be able to do it. Would we want to say that a pigeon is pursuing epistemic rationality when learning to peck a blue rather than red key in order to receive food pellets? There is a sense in which we can say that a pigeon "knows" that the blue key delivers food, but it is not separable from a description of its behaviour which is clearly instrumental. The concept is even less useful in the case of instinctive behaviour. If knowledge is innate it evidently cannot be acquired by the pursuit of epistemic goals. I therefore suggest that epistemic rationality is essentially a concept linked to the new and distinctively human mind.

NEW MIND RATIONALITY

What is fascinating about the human species is both how similar we are to other animals and also how very different. In this section I focus on the kind of human cognition that strongly differentiates us from other species. An alien biologist visiting from a planet of Alpha Centauri would be very

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puzzled indeed by what they observed: a planet with millions of species adapted to their natural environments and hence highly vulnerable to any rapid changes in their surroundings, together with exactly one species that designs, controls, and builds artificial environments and artefacts to suit its own purposes. Not only that, but this one species appears to engage in many activities that have little or no obvious biological purpose, such as writing and reading literature, watching films, and studying ancient history.

While dual process or dual system theory has its critics (see Evans & Stanovich, 2013 for detailed discussion of and reply to various critical articles), the advocates of a second system or new mind are clearly strongly influenced by these unique activities of the human species. Not only is the difference in human achievement from any other species staggering in scale but the thought processes which permit these developments seem qualitatively different. Take engineering design as an example. Old technologies like winches and water wheels were no doubt developed with a large element of trial and error but still must have required much imagination to design. In the modern world an engineer designing a ship or a bridge or an engine will have trained for many years acquiring theoretical knowledge and skills, and will design their artefact with a great deal of hypothetical thinking, far in advance of the product actually being built. Granted, this process will be assisted by study of related previous designs, and a whole range of computational tools and simulations, but the key point is that the design is a *mental representation* well before it is a physical reality. Genetic programming or associative and procedural learning can contribute little to this. This kind of activity is impossible without the full resources of the new mind.

I will not dwell here on how the human species uniquely evolved these extraordinary cognitive powers (but see Evans, 2010; Stanovich, 2004) and will rather focus on their nature and the implications for human rationality. As Stanovich (2013) has recently pointed out, humans may often appear to be less rational in an instrumental sense than other animals. This is because research in judgement and decision making assesses rationality with reference to achieving goals immediately set by the task given to people. Pursuing immediate goals is what animals naturally do. People may appear to lack instrumental rationality when they fail to do so. An example, discussed by Stanovich, is the attention paid to "sunk costs" (e.g., Tan & Yates, 1995) in decision making. If I have expensive tickets bought months ago for an opera tonight but (a) am feeling tired and (b) discover that my favourite football team is playing live on television, then at this point I might prefer to stay home rather than go the opera. But I go to the opera anyway as I have bought the tickets. This is traditionally regarded as irrational because the cost of the tickets is "sunk" and cannot be recovered. It is constant in both choices and so should not affect my decision.

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As Stanovich discusses, phenomena like sunk costs are a much more complex issue because humans have higher-order goals, values, and self-perceptions all of which can affect this kind of decision making. The cost may be sunk but we know we will feel regret if we do not attend. We do not like to waste money or, at a higher level, we do not like to think of ourselves as a person who wastes money, and so on. These complications seem quite obviously to be a consequence of the fact that we have a new mind. An animal lacking such higher-order concepts and representations will naturally follow the path to the immediate goals. Humans pursue all kinds of goals in parallel, and sometimes in conflict, including what may be very long-term objectives such as developing a career or maintaining a relationship. Many of these goals are epistemic, in that we want to discover some knowledge or understand something. While such epistemic goals are often linked to instrumental purpose (studying law in the hope of a well-paid career, for example) they often are not. Why, for example, are people such avid followers of the daily news and trawlers of the Internet when much of the information gleaned will have no impact at all on their actual lives?

It is evident that the new mind is an extraordinary resource that allows us to do the things that other animals cannot do. It does contribute to instrumental rationality in the sense of helping us to achieve our personal goals, including many goals that could not exist without a new mind, such as the goal to travel somewhere we have never seen but only read about. We can nowadays complete all necessary arrangements for such a purpose by applying our new minds to a web browser and without ever leaving our desks! In the process many epistemic goals will be pursued, such as finding good but affordable accommodation, checking the best routes of travel, and discovering whether the climate and facilities of our destination suit our purposes. We will also engage in mental simulations to try to anticipate our experience of the details: Will we be too tired to go out on the first evening? Will the weather be warm enough for us to enjoy swimming? Will a better hotel room feel good enough to justify the extra cost?

My claim is that epistemic rationality is a concept only applicable to the new mind. But are we epistemically rational in the classical sense of seeking true beliefs? In the Internet search example for our holiday, we would certainly want information to be reliable and might seek to confirm this from more than one source. It is also true that many of our new mind plans are successfully executed, implying that the knowledge with which they operate is fit for purpose. Usually, when we plan a holiday to a novel destination, we will actually end up there on the correct date, unless events beyond our control intervene. So we *can* be epistemically rational. But, as mentioned earlier, we can also acquire and maintain false or objectively unjustified beliefs through processes of social influence. Such beliefs do, however, serve an instrumental purpose. For example, a small religious sect might provide a

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number of social reinforcers for adherence to their idiosyncratic dogma. People often feel secure and happy within groups bonded by common belief systems, however ill-founded.

It also appears that we pursue many epistemic goals as apparent ends in themselves, without any goals that are instrumental in any clear biological sense. We follow the news for its own sake, compulsively watch soap operas on television, attempt to solve crossword puzzles, gather enormous amounts of information about our favourite sports team, and so on. And how is it "truth seeking" to enjoy watching supernatural movies or reading science fiction? It is evident that humans enjoy a rich mental life in which curiosity and imagination play a large part. While the basic biological drivers of survival and reproduction remain, much of our modern lives takes such basics for granted, at least in the wealthier parts of the world. The bulk of our instrumental and epistemic goals are set by the complex social, economic, and political structures only made possible by the development of our new minds.

IRRATIONALITY AND TWO MINDS CONFLICT

My assumption is that instrumental rationality is fundamental and epistemic rationality only a means to its end. I have argued that the old mind seeks to repeat past success by mechanisms that implicitly code behaviour patterns. For this reason, I see epistemic rationality as only usefully applying to the new mind, which explicitly processes knowledge in order to decide on future actions. To deem that someone is *irrational* in the instrumental sense is to say that they act in such a way as not to achieve their goals. To say that they are irrational in the epistemic sense is to say that they hold false beliefs or reason incorrectly, with the *consequence* that they fail to achieve their goals.

In practice, attributions of irrationality are very complex and difficult to make, and arguably it is no business of the psychologist to make them, certainly in the domain of basic research into cognitive mechanisms (Elqayam & Evans, 2011). However, since we study motivated behaviour, some notion of instrumentality is inescapable. In the study of judgement and decision making (JDM), attributions of irrationality have largely concerned deviations from normative principles, with observed judgements often failing to conform to behavioural decision theory and probability theory. In the psychology of reasoning, norms are again applied but more in the domain of epistemic rationality, with claims that people fail to reason logically or are influenced by irrelevant beliefs and other biasing factors. These studies have produced a rich source of data about human thought processes and I have myself written several books about the cognitive biases observed in these fields (most recently, Evans, 2007a). However, as discussed above, the notion that norm violation necessarily indicates irrationality has proven

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highly controversial and the main cause of the Great Rationality Debate from Cohen (1981) onwards.

Putting aside normative rationality, I wish here to add some observations on the issue of irrationality as viewed from the perspective of the two minds theory. The theory proposes that the two minds may pursue different goals by different mechanisms. The problem of goal conflict in a dual system framework has been discussed in detail by Stanovich (2004) who highlights the distinction between evolutionary and individual rationality. He suggests that System 1 (old mind) is driven by the former and System 2 (new mind) by the latter, creating a conflict which he sees as the primary cause of the many cognitive biases reported in studies of decision making and reasoning. I partly agree with this, but see the formation of habits in the old mind by its general learning mechanism as another major cause of two minds conflicts and cognitive biases. For example, when we make decisions out of habit, we may overlook novel features in a problem which would require new mind resources for its solution.

In general, the two minds work well in concert, so that low-effort processing takes care of routine background tasks while new mind resources are devoted to that which is novel and demanding of controlled attention. But two minds conflict clearly does occur. It can be relatively harmless, for example when forgetting to deviate from one's habitual route of driving to work, despite having planned a stop on the way. It can also have much more serious consequences, as when a compulsive gambling habit can destroy an individual's life. Another example would be a severe anxiety problem which obstructs an individual from carrying their person goals, such as a lift phobia which might prevent someone attending a critical career interview on a high floor of a skyscraper. In such cases the individual is normally aware of the conflict and may be highly distressed by it. From the point of view of the individual's goals, as represented in the new mind, these behaviours may appear irrational to others and the individual themselves. Such judgements of irrationality are, however, subjective and a reflection of social attitudes. Those who smoke or who are heavily obese often receive little sympathy from others when they experience severe health consequences of their condition. The assumption of the non-smoker or thin person seems to be that the individuals in question could choose not to smoke or overeat and are therefore responsible for their own problems. To me, the evidence suggests that, in two minds conflict, the old mind very often dominates and it is not at all easy to impose the new mind upon it. This seems clear from the grip that phobias, compulsive behaviours, and behavioural addictions such as gambling can exert on individuals to the detriment of their lives, and the great difficulty they have in freeing themselves from these manifestations of the old mind.

The folk psychological notion of irrationality is that the *person* fails to act in such a way as to achieve their goals. But the person is not the

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organism. It is a construction of the new mind that contains an illusion of control and free will (see, for example, Bargh & Ferguson, 2000; Wegner, 2002; Wilson, 2002; for detailed discussion of this issue see Evans, 2010, Ch. 7). The constructed self with its illusory control is an important part of the two minds theory. So, in folk psychology, we feel ourselves and others to be irrational when we or they act in such a way as to bring about bad consequences: an impulsive act of dishonesty which is found out; a disastrous sexual adventure; a rash investment that loses money, and so on. In reality we might have had little choice in the matter! On a two minds view, such apparently irrational acts can often be seen as reflecting old mind influences with emotions such as lust and greed taking control, or habits and intuitions exerting themselves when high-effort thinking and reasoning was required. Hence old mind mechanisms, which evolved through its own form of instrumental rationality, may have the effect of creating irrational behaviour when viewed from the perspective of the person. But the old mind is part of us, viewed as an organism, and the person may not operate without it.

CONCLUSIONS

The two minds theory has major implications for our views on human rationality. First, it is clear that while both minds strive for instrumental rationality—the achievement of goals—they have very different mechanisms for achieving this. The old mind relies on a combination of evolved cognitive systems and mostly, in the case of humans, experiential learning in order to repeat behaviours that have been successful in the past. This is a form of instrumental rationality which we share with other animals. The new mind, by contrast, can seek to achieve goals by imagining the future, engaging in mental simulations, and consequential decision making. This enables humans uniquely to decide the best course of action with novel problems for which experiential learning can provide no solutions. Hence this can provide a mechanism for instrumental rationality which is not available to other animals. For this reason I have also argued that epistemic rationality—the attainment of true, or fit for purpose, knowledge—is only a useful concept when talking about the new and distinctively human mind. The reasoning of the new mind relies on access to explicit knowledge systems whose accuracy may be critical for solving problems.

In my discussions of two minds rationality, I have constantly referred to the manner in which the two minds *strive* to achieve goals based on the cognitive capability we have so far managed to evolve. It is the striving that to me is axiomatic, not the actual achievement. "Irrationality" is a highly emotive and pejorative term, and its attribution to multiple examples of norm violation in the literatures on reasoning and decision making has evoked much heated debate and consumed many pages of learned journals. The

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point of such experiments was never to show that people behaved in a manner that was stupid, irresponsible, and reckless. Nor should it matter that such experiments are often unrepresentative of real-life situations (Cohen, 1981)—so are studies of the serial position curve in memory research. The purpose is to discover the mechanisms that underlie our motivate cognition and their limitations.

It is important to understand the different mechanisms employed by the two minds to achieve their goals, and to understand the conflict between goals that may occur in our complex cognitive architecture. This is equally true for the cognitive theorists and for those whose objective is to study limitations in cognitive performance—by some objective standard—in order to improve it by education and training. We need, for example, to know when training in explicit rule-based reasoning will suffice and when more indirect approaches are needed to deal with cognitive biases. However, we do not need to adopt the rhetoric of irrationality to achieve this purpose, and in my view would be much better off avoiding it.

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